

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed **predictive analytics models** using **Python (pandas, scikit-learn, NumPy)** and **statistical hypothesis testing** to identify payment transaction anomalies across 2.5M+ daily SWIFT messages, reducing false-positive incident alerts by **32%** and improving SLA compliance to **99.2%**.
- Built **data pipelines and ETL workflows** using **Python automation** and **SQL (PostgreSQL, Oracle, MySQL)** to consolidate enterprise banking operations data across distributed systems; engineered **scheduled reporting jobs** via **Linux cron** that delivered real-time KPI dashboards to **150+ operations stakeholders** daily.
- Designed and deployed **Power BI dashboards and DAX measures** for payments-operations monitoring, applying **data modelling** best practices to visualize **ISO 20022 message flows, SEPA reconciliation status, and cross-border payment SLAs**; dashboards tracked **€450M+ in daily transaction volume** with **sub-second refresh**.
- Conducted **exploratory data analysis (EDA)** and **time-series forecasting** using **R/RStudio and scikit-learn** to forecast payment processing latency and operational capacity; delivered statistical insights that informed resource-allocation decisions, reducing peak-hour processing delays by **28%**.
- Engineered **cloud-based data warehouse integrations** on **AWS (EC2, S3, Lambda)** using **Docker containerization** and **Jenkins CI/CD pipelines** to ingest, validate, and transform banking domain datasets; implemented **Git branching and code-review workflows** ensuring data quality and regulatory compliance (**GDPR / audit trails**) across **12+ interconnected enterprise systems**.
- Applied **machine learning evaluation metrics (precision, recall, ROC-AUC)** and **A/B testing methodologies** to assess anomaly-detection models against production payment data, achieving **94.7% accuracy** in identifying fraudulent SWIFT transactions; documented findings in executive-ready **Tableau interactive dashboards** with **business intelligence KPIs** for C-suite stakeholders.

SKILLS

Data Science & Analytics – predictive modeling, statistical hypothesis testing, exploratory data analysis, A/B testing, time-series forecasting, machine learning (supervised/unsupervised), evaluation metrics (precision, recall, ROC-AUC, F1), scikit-learn, scikit-optimize.

Programming & Data Engineering – Python (pandas, NumPy, scikit-learn), R/RStudio, SQL (PostgreSQL, Oracle, MySQL), ETL pipeline development, data validation, scheduled reporting, shell scripting, Linux.

Business Intelligence & Visualization – Power BI (dashboards, DAX measures, Power Query, data modelling), Tableau (interactive dashboards), data warehousing, data governance, regulatory reporting (GDPR, audit trails).

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins CI/CD pipelines, Git, code review, change-control management.

Domain Expertise – payments processing, SWIFT messaging (MT/MX formats), ISO 20022 message flows, SEPA payments, cross-border payments, PSD2 / Open Banking, banking SLAs, financial regulatory compliance, ERP integration.

Soft Skills & Methodologies – Agile/Scrum, cross-functional stakeholder collaboration, UAT coordination, production support, incident management, data-driven decision-making, technical documentation.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Real-Time Payment Anomaly Detection & Predictive Analytics Platform (HCL Technologies)

- Architected an **end-to-end machine learning pipeline** ingesting **2.5M+ daily SWIFT payment messages** via **AWS Lambda** and **S3**, applying **scikit-learn classification models** (Random Forest, Gradient Boosting) to detect fraudulent transactions and operational anomalies in real time.
- Engineered **feature engineering and EDA workflows** using **pandas and NumPy** to extract temporal, behavioral, and transactional signals from **ISO 20022 message structures**; performed **A/B testing** on model variants, achieving **94.7% precision and 91.3% recall** on held-out SWIFT test sets.
- Deployed **Power BI dashboards with DAX measures** and **Tableau interactive visualizations** to surface model predictions, anomaly scores, and **SEPA reconciliation KPIs** to **120+ banking operations teams**; integrated **PostgreSQL data warehouse** queries for **sub-second refresh rates** on **€450M+ daily payment volume**.
- Implemented **data pipeline orchestration** using **Python scripts, Docker containers, and Jenkins CI/CD**; automated **daily retraining, model validation, and performance monitoring** via **Linux cron jobs**, ensuring **99.2% SLA compliance** and regulatory-grade **GDPR audit trails**.
- Conducted **time-series forecasting** using **R statistical methods** to predict payment-processing latency during peak hours; delivered **executive-ready reports** that informed capacity-planning decisions, reducing operational delays by **28%** and improving customer experience across **PSD2-compliant channels**.

SWIFT Payments Operations Dashboard & Data Governance Framework (HCL Technologies)

- Designed and operationalized a **Power BI payments-operations dashboard** consolidating real-time **SWIFT MT/MX message flows, ISO 20022 transaction status, SEPA payment reconciliation, and cross-border payment SLAs** across **6 banking-domain systems** for **C-suite and operations leadership**.
- Built **ETL data pipelines** using **SQL (PostgreSQL, Oracle) and Python automation** to extract, validate, and normalize payment data from interconnected enterprise systems; implemented **change-control management** protocols ensuring **zero-downtime deployments** during monthly UAT cycles.
- Established **data governance and quality assurance frameworks** with **GDPR compliance checks, audit logging, and regulatory reporting**; documented **data lineage and transformation logic** in **Confluence** for **15+ cross-functional stakeholders** (BAs, operations, compliance teams).

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)